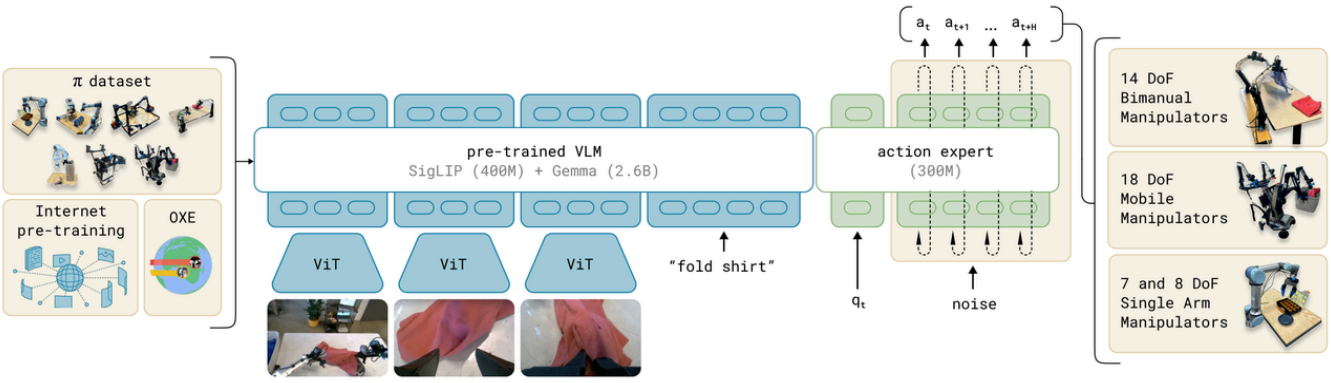
π0: A Vision-Language-Action Flow Model for General Robot Control

> #VLA #HybridArchitecture



- What problems does this architecture solve?
  - Currently, the robotics area faces 3 main problems: poor generalization ability, lack of data, and low robustness. The previous work was only for a specific type of robot.
  - This project is to find a model such that it can run on various platforms and is able to output consecutive high-precision actions.
- How does this architecture work?
  - Architecture & Inference
    - This architecture uses a pretrained VLM based on a large set of data. The VLM will take and process the inputs: multi-view images and NL descriptions, making the model gain powerful capabilities of handling common-sense and semantics.
    - Images and NL prompts will be converted to embeddings, and self-attention will be performed.
    - There is a "Motor Cortex", an expert with a size of 300M parameters. It will receive the robot's physical status and action vectors with noise. The noisy action vector will use attention to attend the processed images, processed NL, and the physical status. **Note that this is single-directional attention, meaning that the NL-IMG-STAT parts cannot attend to the noisy actions.** This will ensure that the pretrained common-sense and visual representations will not be contaminated by the robot-specific signals.
    - The model will predict H steps of action ($H = 50$ in the paper). It will, like normal flow-match models, use **Forward Euler Integral** to denoise, recovering the action vector.

      $A_t^{\tau+\delta} = A_t^\tau + \delta v_\theta(A_t^\tau, o_t)$
    - For a very complex task, a powerful VLM will be used to split the task into subtasks, which will be then fed to π_0.
  - Training and Finetuning
    - Objective function: MSE of the speed, like those in flow match models:

$$L^{\tau}(\theta) = \mathbb{E}_{p(A_t|o_t),q(A_t^{\tau}|A_t)}||v_\theta(A_t^{\tau}, o_t) - u(A_t^{\tau}|A_t)||^2, \text{where}$$
$$o_t : \text{All observations: image, NL, physical status}$$
$$A : \text{Action vector}$$

    - Pretraining: Pretrain the model with a large number of topics on various topics, covering 7 types of robot and 68 types of tasks.
    - Post-training: Uses small-scale but highly curated, high-quality downstream task data.
- Experiment Results
    - Excellent out-of-the-box performance: Even without fine-tuning for specific tasks, the pre-trained π_0 model significantly outperforms previous baseline models such as OpenVLA and Octo on zero-shot tasks.
    - Tackling complex tasks with extremely long durations: After fine-tuning, π_0 can complete complex multi-stage tasks ranging from several minutes to twenty minutes in duration (such as folding clothes in a washing machine, clearing a table filled with unknown objects, and assembling cardboard boxes).
    - The value of pre-training: Experiments demonstrate that for more challenging tasks (such as novel operations not seen in the task library), pre-trained models deliver more significant performance gains. This not only outperforms task-specific models trained from scratch but also validates the feasibility of a general-purpose robotics foundation.
- Work Cited
    - Paper
    - Chat History
    - NotebookLM